

AKASH UPADHYAY

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Professional Summary

Mechanical Engineering graduate with a self-driven transition into software development, having built and shipped full-stack projects using Next.js, React, Node.js, and TypeScript. Strong foundation in data structures and algorithms. Currently completing an Engineer Trainee program at Infosys, including training in AI awareness, while independently continuing to build software skills through project work.

Education

Sant Longowal Institute of Engineering and Technology

Nov 2022 – May 2026

B.E. Mechanical Engineering — Current GPA: 8.49/10.0

- Relevant Coursework: Data Structures & Algorithms, Web Development

Skills

Languages: C++, JavaScript, TypeScript, SQL, Motoko

Web Frameworks/Libraries: Next.js, React, Node.js, Express.js, Tailwind CSS

Databases/ORMs: MySQL

Tools/Platforms: Git, GitHub, VS Code, Postman

Development Practices: Monorepo architecture (Turborepo), RESTful API design, Git version control

Projects

Payment Transfer Web Application | *Next.js, TypeScript, NextAuth.js, Tailwind CSS, Turborepo*

- Developed a full-stack payment platform replicating core Paytm functionalities, supporting wallet funding, peer-to-peer transfers, and secure transaction management using Next.js and Node.js.
- Implemented row-level locking for wallet transactions, preventing race conditions and double spending during concurrent balance updates.
- Created a custom webhook simulation framework processing 100+ real-time payment events per minute, enabling independent end-to-end testing without third-party dependencies.
- Engineered RESTful API endpoints supporting asynchronous transaction updates while maintaining data consistency and reliable payment state synchronization across services.
- Styled 15+ reusable Tailwind CSS UI components, delivering responsive experiences across desktop and mobile platforms.
- Organized the project using a Turborepo monorepo structure, reducing build/deployment time and improving code reusability and maintainability.

Decentralised Token Application | *JavaScript, Node.js, React.js, Tailwind CSS, Motoko, DFINITY SDK (dfx)*

- Developed a token canister in Motoko, enabling creation and transfer of up to 1,000,000,000 tokens with persistent state storage on the Internet Computer.
- Integrated a React.js frontend with a Motoko-based backend canister using DFINITY APIs, enabling 4 blockchain operations including token transfer, balance queries, token distribution, and asset management.
- Implemented 3 core token management workflows — faucet distribution, balance retrieval, and token transfers — while maintaining consistent ledger state during concurrent user interactions.
- Architected a responsive React.js frontend with 5 reusable components using Tailwind CSS, enabling intuitive and maintainable token management workflows.

Training & Experience

Infosys Limited — Engineer Trainee

Current

- Undergoing a structured trainee program covering core mechanical engineering skills, along with training in AI awareness and emerging technologies.
- In parallel, continuing self-directed software development learning outside the core curriculum — solving Data Structures and Algorithms problems across LeetCode, GeeksforGeeks, and CodeStudio to strengthen algorithmic proficiency.
- Independently applying these self-taught software skills to full-stack project work, sustaining hands-on development experience alongside the trainee program.

Extracurriculars

- Organized and led voter awareness drives through the SLIET Electoral Literacy Club, engaging hundreds of students across campus.
- Ranked among top finalists in a State-level Spell Bee competition.
- Competed in 10+ inter-school and inter-college debate competitions, honing public speaking and analytical thinking.